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The range of Excess-8 code is from _____ to _____.

Answer (Please select your correct option)

☐ +7 to -8
correct

☐ +8 to -7

☐ +9 to -8

☐ -9 to +8



The decimal "17" in BCD will be represented as _____

Answer (Please select your correct option)

☐ 11101

☐ 11011

☐ 10111

☐ 11110

correct

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The four outputs of two 4-input multiplexers, connected to form a 16-input multiplexer, are connected together through a 4-input _____ gate

Answer (Please select your correct option)



AND



OR



NAND



XOR

correct

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A standard interface for programming the In-System PLD consists of _____.

Answer (Please select your correct option)



2-wire



4-wire

correct



8-wire



16-wire

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_____ Dual, 4-input multiplexer can be connected to form a 16-input multiplexer.

Answer (Please select your correct option)

☐

2

☐

3

☒

4

correct

☐

5

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S-R latch can be implemented by using _____ gates

Answer (Please select your correct option)

☐ AND, OR

correct

☐ NAND, NOR

☐ NAND, XOR

☐ NOT, XOR

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A product term is 0 when_____.

Answer (Please select your correct option)



Any of the literals is 1



At least two literals are 1



All the literals are 1



Any of the literal is 0

correct

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What will be the output of a 3-input NAND gate for the input values: A=1, B=0, C=1

Answer (Please select your correct option)

☐ Zero

correct

☐ One

☐ Undefined

☐ No output, as input is invalid

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_____ is not a valid hexadecimal number.

Answer (Please select your correct option)



1234



ABCD



1001



DEHF

correct

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2's complement of any binary number can be calculated by

Answer (Please select your correct option)

☐ adding 1 to 1's complement

correct

☐ subtracting 1 from 1's complement.

☐ calculating 1's complement and inverting Most significant bit

☐ adding 1's complement twice

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A BCD to 7-Segment decoder has _____ inputs and _____ outputs.

Answer (Please select your correct option)

☐ 3, 7

☐ 4, 7

correct

☐ 7, 3

☐ 7, 4

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Which of the following simplification method is intended to be used for Boolean expressions having more than four variables?

Answer (Please select your correct option)

☐ Boolean Algebra and rules

☐ Karnaugh Map

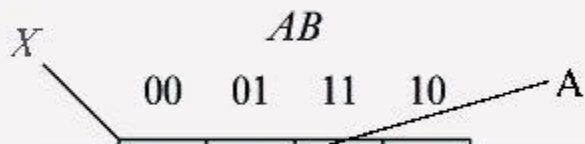
☒ Quine-McCluskey

correct

☐ Demorgan's Theorem

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In the following Karnaugh map, which group has "legal grouping" ?



Answer (Please select your correct option)

A

☐

B

☐

C

☒

correct

D

☐

CD

00	1	1	1	0
01	1	0	0	1
11	0	1	1	1

B

C

D

Answer (Please select your correct option)

☐ A☐ B☐ C☐ D

11	0	1	1	1
10	0	1	1	0

D

Answer (Please select your correct option)

☐ A☐ B☐ C☐ D



What are the values of sum and carry out when two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder and carry input is 1.

Answer (Please select your correct option)

☐ $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 0111, C_{out} = 0$

☐ $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1111, C_{out} = 1$

☐ $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1011, C_{out} = 1$

☐ $\Sigma_4 \Sigma_3 \Sigma_2 \Sigma_1 = 1100, C_{out} = 1$



A particular half-adder has _____ inputs and _____ output(s).

Answer (Please select your correct option)

☐ 3, 1

☐ 3, 2

☐ 2, 1

☐ 2, 2

correct

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TTL based devices work with a DC supply of ____ Volts

Answer (Please select your correct option)

+10



+5



correct

+3



3.3



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In CMOS 5 Volt series, Input voltage for Logic high signal (V_{IH}) is in the range of _____ volts.

Answer (Please select your correct option)

☐ 3.5 to 5

correct

☐ 4.5 to 5

☐ 0 to 5

☐ 0 to 3.5

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A multiplexer circuit has _____ input(s) and _____ output(s).

Answer (Please select your correct option)

☐ Single, single

☐ Single, multiple

☐ Multiple, single

☐ Multiple, multiple

correct

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The binary values for the standard SOP expression, $ABC\bar{D} + \bar{A}BC\bar{D} + \bar{A}\bar{B}\bar{C}D$ are _____

Answer (Please select your correct option)

☐ 1110 + 0110 + 0001

correct

☐ 1011 + 1111 + 1011

☐ 0001 + 1001 + 1110

☐ 1010 + 1110 + 0101

Determine the binary values of the variables for the following standard POS expression.

$$(A + \bar{B} + C)(\bar{A} + B + \bar{C})$$

Answer (Please select your correct option)

☐ $(1 + 0 + 1)(0 + 1 + 0)$

correct

☐ $(0 + 1 + 0)(1 + 0 + 1)$

☐ $(1 + 0 + 0)(0 + 0 + 1)$

☐ $(0 + 1 + 0)(1 + 1 + 0)$



The first Least Significant digit in decimal number system has

Answer (Please select your correct option)



position 0 and weight equal to 1

correct



position 1 and weight equal to 0



position 1 and weight equal to 10



position 0 and weight equal to 10

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Consider the binary number "100101", its 2's complement will be _____

Answer (Please select your correct option)



100110



011011

correct



100111



011010

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$$A + \overline{A}.B = \underline{\hspace{2cm}}$$

Answer (Please select your correct option)

☐ A

☐ B

☐ A.B

☐ A+B

According to Demorgan's theorem:

$$\overline{A + B + C} = \underline{\hspace{2cm}}$$

Answer (Please select your correct option)

☐ $A.B.C$

☐ $A + \overline{B.C}$

☐ $\overline{A.B.C}$

correct

☐ $\overline{A.B} + C$

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The Extended ASCII Code (American Standard Code for Information Interchange) represents _____ unique codes

Answer (Please select your correct option)

256



correct

255



128



127



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The three fundamental gates are _____

Answer (Please select your correct option)

☐

AND, NAND, XOR

☐

OR, AND, NAND

☐

NOT, NOR, XOR

☐

NOT, OR, AND

correct

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For a Standard SOP expression, a _____ is placed in the cell corresponding to the product term (Minterm) present in the expression.

Answer (Please select your correct option)

0



1



correct

X (don't care condition)



Any of given option depending on SOP term.



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The output $A < B$ is set to 1 when the input combinations is _____

Answer (Please select your correct option)

☐ A=10, B=01

☐ A=11, B=01

☐ A=01, B=01

☐ A=01, B=10

correct

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An alternate method of implementing Comparators which allows the Comparators to be easily cascaded without the need for extra logic gates is _____

Answer (Please select your correct option)

- ☐ Using a single comparator
- ☐ Using Iterative Circuit based Comparators
- ☐ Connecting comparators in vertical hierarchy
- ☐ Extra logic gates are always required.

correct



Two 2-input, 4-bit multiplexers 74X157 can be connected to implement a ____ multiplexer.

Answer (Please select your correct option)

☐ 2-input, 8-bit

correct

☐ 2-input, 4-bit

☐ 4-input, 8-bit

☐ 4-input, 16-bit



$A.(B.C) = (A.B).C$ is an expression of _____

Answer (Please select your correct option)



Demorgan's Law



Distributive Law



Commutative Law



Associative Law

correct

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$(A + B)(A + \bar{B} + C)(\bar{A} + C)$ is an example of _____

Answer (Please select your correct option)



Product of sum form

correct



Sum of product form



Demorgans law



Associative law

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The expression $F=A+B+C$ describes the operation of 3-bit _____ gate.

Answer (Please select your correct option)

OR



correct

AND



NOT



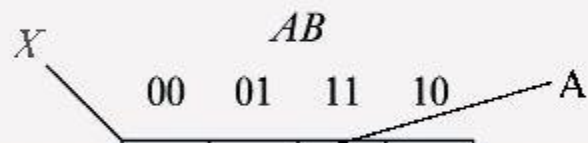
NAND



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In the following Karnaugh map, which group has "legal grouping" ?



Answer (Please select your correct option)

A

☐

B

☐

C

☒

correct

D

☐



A particular Full Adder has _____ inputs and _____ output(s).

Answer (Please select your correct option)

☐ 2, 3

☐ 2, 2

☐ 3, 2

correct

☐ 3, 3

CD

00	1	1	1	0
01	1	0	0	1
11	0	1	1	1

B

C

Answer (Please select your correct option)

A

☐

B

☐

C

☒correct

D

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CD

11

0

1

1

1

10

0

1

1

0

D

Answer (Please select your correct option)

A

☐

B

☐

C

☒correct

D

☐



Which of the following is a symbol of OR operation in ABEL?

Answer (Please select your correct option)

#



correct

\$



!



&



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A latch has _____ stable state(s).

Answer (Please select your correct option)

- ☐ One
- ☒ Two correct
- ☐ Three
- ☐ Four

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A latch retains its state unless _____

Answer (Please select your correct option)



Power is turned off



Input is changed

correct



Output is changed



Clock pulse is changed

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At $S = 0$ and $R = 1$, an active-HIGH SR latch is in _____ condition.

Answer (Please select your correct option)



SET



RESET



Invalid



No change

correct

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What are the sum (Σ) and the carry (C_{out}) for full adder circuit when $A = 1$, $B = 1$ and $C_{in} = 0$?

Answer (Please select your correct option)

$\Sigma = 0, C_{out} = 1$



correct

$\Sigma = 1, C_{out} = 0$



$\Sigma = 1, C_{out} = 1$



$\Sigma = 0, C_{out} = 0$



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Consider the binary number "100101", its 2's complement will be _____

Answer (Please select your correct option)



100110



011011

correct



100111



011010

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The values that exceed the specified range can not be correctly represented and are considered as _____

Answer (Please select your correct option)

☐ Overflow

correct

☐ Carry

☐ Parity

☐ Sign value

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According to Demorgan's theorem:

$$\overline{A + B + C} = \underline{\hspace{2cm}}$$

Answer (Please select your correct option)

☐ $A.B.C$

☐ $A + \overline{B.C}$

☐ $\overline{A.B.C}$

correct

☐ $\overline{A.B} + C$

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If two numbers in BCD representation generate an invalid BCD number then the binary _____ is added to the result

Answer (Please select your correct option)



1001



0110



1111



1100

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The output of an AND gate is one when _____.

Answer (Please select your correct option)



All of the inputs are one

correct



Any of the input is one



Any of the input is zero



All the inputs are zero

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A standard SOP form has _____ terms that have all the variables in the domain of the expression.

Answer (Please select your correct option)



Sum



Product

correct



Max



Composite

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A comparator circuit has multiple inputs and _____ outputs

Answer (Please select your correct option)

☐ 2

☐ 3

correct

☐ 4

☐ 7

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An alternate method of implementing Comparators which allows the Comparators to be easily cascaded without the need for extra logic gates is _____

Answer (Please select your correct option)

- ☐ Using a single comparator
- ☐ Using Iterative Circuit based Comparators
- ☐ Connecting comparators in vertical hierarchy
- ☐ Extra logic gates are always required.

correct

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The active high and active low inputs of 3-to-8 Decoder are as follows:

Answer (Please select your correct option)



One active-high and the remaining two are active-low.

correct



Two active-high and the remaining one is active-low.



All active high



All active low

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OLMC consists of a _____

Answer (Please select your correct option)



Encoder



Decoder



Logic Gate



Flip-Flop

correct

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The Quad Multiplexer has _____ inputs

Answer (Please select your correct option)

4



8



12



16



correct

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The carry propagation can be expressed as _____.

Answer (Please select your correct option)



$$C_p = AB$$



$$C_p = A + B$$



$$C_p = A \oplus B$$



$$C_p = A + \bar{B}$$

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In the binary number "1100.111", the first bit at the right of decimal point has the weight

Answer (Please select your correct option)

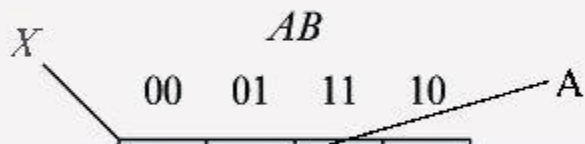
☐ $\frac{1}{2}$ (1 divided by 2)

☐ $\frac{1}{4}$ (1 divided by 4)

☐ 0

☐ 1

In the following Karnaugh map, which group has "legal grouping" ?



Answer (Please select your correct option)

A

☐

B

☐

C

☒

correct

D

☐

In CMOS 5 Volt series, Input voltage for Logic high signal (V_{IH}) is in the range of _____ volts.

Answer (Please select your correct option)

☐ 3.5 to 5

correct

☐ 4.5 to 5

☐ 0 to 5

☐ 0 to 3.5

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Programmable Array Logic (PAL) has _____ input(s) and _____ output(s).

Answer (Please select your correct option)

☐ Single, multiple

☐ Multiple, single

☐ Single, single

☐ Multiple, multiple

correct

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An equivalent ABEL notation of Boolean expression **A+B** is _____

Answer (Please select your correct option)

☐ A & B

☐ A I B

☐ A # B

correct

☐ A \$ B

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Which circuit has the following Sum and CarryOut expressions?

$$\text{Sum} = A \oplus B \oplus C$$

$$\text{CarryOut} = C(A \oplus B) + AB$$

Answer (Please select your correct option)

☐ Half Adder

☒ Full Adder

☐ 3-bit parallel adder

☐ MSI adder circuit

correct

From the truth table below, determine the standard SOP expression.

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0

Answer (Please select your correct option)

☐

$$\overline{A}\overline{B}C + \overline{A}BC + A\overline{B}\overline{C}$$

☐

$$AB\overline{C} + \overline{A}B\overline{C} + \overline{A}\overline{B}C$$

correct

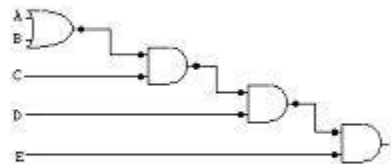
☐

$$(A + B + \overline{C})(A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

☐

$$(\overline{A} + \overline{B} + C)(\overline{A} + B + C) + (A + B + \overline{C})$$

The Boolean expression for the logic circuit is _____



Answer (Please select your correct option)

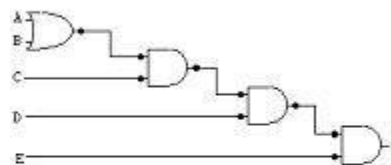
☐ $AB+CDE$

☐ $(A+B)CDE$

correct

☐ $(A+B)+(CDE)$

☐ $(A+B)(CD+E)$



Answer (Please select your correct option)

☐ $AB+CDE$

☐ $(A+B)CDE$

correct

☐ $(A+B)+(CDE)$

☐ $(A+B)(CD+E)$

Made By: Waqar Siddhu



$(A+B).(A+C) =$ _____

Answer (Please select your correct option)

☐

$B+C$

☐

$A+BC$

correct

☐

$AB+C$

☐

$AC+B$

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Excess-8 code assigns _____ to "-8"

Answer (Please select your correct option)

☐ 1110

☐ 1100

☐ 1000

☐ 0000

correct

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The output of the expression $F = A.B.C$ will be Logic _____ when $A=1$, $B=0$, $C=1$.

Answer (Please select your correct option)

☐ Undefined

☐ One

☒ Zero

correct

☐ No Output as input is invalid.



The cell marked 6 in 4-variable K-Map represent minterm 6 or the maxterm 6 having the following binary value of variables A, B, C and D.

Answer (Please select your correct option)

☐

A=1, B=1, C=0, D=0

☒

A=0, B=1, C=1, D=0

correct

☐

A=0, B=0, C=1, D=1

☐

A=1, B=0, C=0, D=1

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A comparator circuit has multiple inputs and _____ outputs

Answer (Please select your correct option)

☐ 2

☐ 3

correct

☐ 4

☐ 7

The output $A < B$ is set to 1 when the input combinations is _____

Answer (Please select your correct option)

☐

A=10, B=01

☐

A=11, B=01

☐

A=01, B=01

☐

A=01, B=10

correct

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The _____ Encoder is used as a keypad encoder.

Answer (Please select your correct option)

☐

2-to-8 encoder

☐

4-to-16 encoder

☐

BCD-to-Decimal

☐

Decimal-to-BCD Priority

correct

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Using multiplexer as parallel to serial converter requires _____ connected to the multiplexer

Answer (Please select your correct option)



A parallel to serial converter circuit

correct



A counter circuit



A BCD to Decimal decoder



A 2-to-8 bit decoder



The four outputs of two 4-input multiplexers, connected to form a 16-input multiplexer, are connected together through a 4-input _____ gate

Answer (Please select your correct option)

☐

AND

☒

OR

correct

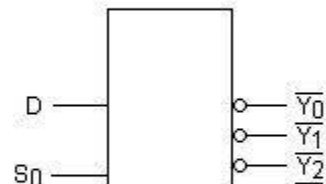
☐

NAND

☐

XOR

The device shown here is most likely a _____.



Answer (Please select your correct option)

☐ Comparator

☐ Multiplexer

correct

☐ Demultiplexer

☐ Parity generator



GAL is an acronym for _____.

Answer (Please select your correct option)



Generic Analysis Logic



Giant Array Logic



General Array Logic



Generic Array Logic

correct

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In the binary number "10011" the weight of the most significant digit is ____.

Answer (Please select your correct option)



2^4 (2 raise to power 4)

correct



2^3 (2 raise to power 3)



2^0 (2 raise to power 0)



2^1 (2 raise to power 1)



The output of the expression $F=A+B+C$ will be Logic _____ when $A=0$, $B=1$, $C=1$.

Answer (Please select your correct option)

☐ Undefined

☐ One

☐ Zero

☐ 10 (binary)

correct

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Which of the following simplification method is intended to be used for Boolean expressions having more than four variables?

Answer (Please select your correct option)

☐ Boolean Algebra and rules

☐ Karnaugh Map

☒ Quine-McCluskey

correct

☐ Demorgan's Theorem



A particular half-adder has _____ inputs and _____ output(s).

Answer (Please select your correct option)

☐ 3, 1

☐ 3, 2

☐ 2, 1

☐ 2, 2

correct

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A quantity having continuous values is a/an _____ quantity.

Answer (Please select your correct option)

☐ digital

☐ binary

☐ analog

correct

☐ natural



Which one of the following expressions is an example of Commutative Law for Multiplication?

Answer (Please select your correct option)

☐

$$AB+C = A+BC$$

☐

$$A(B+C) = B(A+C)$$

☒

$$AB=BA$$

correct

☐

$$A+B=B+A$$

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When four 1's are taken as a group on a Karnaugh map, the number of variables eliminated from the output expression is/are _____

Answer (Please select your correct option)

☐ 1

☐ 2

☐ 3

☐ 4



Identify the Boolean expression that is in standard POS form.

Answer (Please select your correct option)

☐ $(A + \bar{B})(\bar{A} + B + \bar{C})$

correct

☐ $\bar{A}\bar{B}C + A\bar{B}C$

☐ $(A + \bar{B} + \bar{C})(A + B + C)$

☐ $(A + B)(A + C)$



Simplifying POS (Product of sum) expression requires the grouping of _____

Answer (Please select your correct option)



0s

correct



1s



Don't care



Both 0s and 1s